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Kryptos Hints and Clues

from Jim Sanborn and Ed Scheidt

I have included here quotes from Jim Sanborn and Ed Scheidt, that I believe could potentially be clues (or provide important insight). All are direct quotes from them, unless mentioned otherwise (e.g. "(via reporter)").

If time permits at some point, I want to look for more quotes, add keywords (such as "masking"), and make it more "web 2.0" (show/unshow based on the person, source, etc.). And perhaps order them by date or other logical structure.

Jim Sanborn	NPR 09 Jun 2005	And I'm finding that it's increasingly difficult to keep that kind of information secret because the closer people get to cracking the last 97 characters, the more difficult it is to say absolutely nothing about what it does or does not say, and so it's tough.
Jim Sanborn	NPR 09 Jun 2005	I don't really believe I gave them the entire code. So, you know, if it was a deception on my part, hey, so be it. You have to play the game the way everyone else, you know, at the agency plays the game.
Ed Scheidt	NPR 09 Jun 2005	For instance, if you could determine that--the number of E's vs. the number of Z's in the English language, then you would have some insight into the potential words that they made up, because there are more E's in the English language than there are Z's.
Ed Scheidt (via reporter)	NPR 09 Jun 2005	Scheidt says in part four he deliberately masked that advantage. "Kryptos" sleuths speculate that could mean anything from random Q's and X's being thrown in to phonetic spellings such as writing the number four, P-H-O-A-R.
Jim Sanborn	Wired News 20 Jan 2005	The work that happened before the CIA, and (was) the reason I was chosen (to do the CIA piece), dealt with invisible forces -- albeit natural invisible forces. I built pieces about the Coriolis force that makes whirlpools in the Northern and Southern hemispheres turn in opposite directions. I also worked with the Earth's magnetic field and worked with lodestones ... from which we derived our knowledge of electricity. It was all involved in the secrets of nature before the agency chose my work to deal with the secrecy of man.
Jim Sanborn	Wired News 20 Jan 2005	WN: Is it important to look at your other works, before and after Kryptos, to understand Kryptos? Sanborn: For the student of cryptography it's always helpful to gather as much information as possible when zeroing in on and encoding a system.
Jim Sanborn	Wired News 20 Jan 2005	WN: You also have two other sculptures on the CIA grounds that are near the Kryptos sculpture. Are they related to Kryptos in some way? Sanborn: Well, all I'll say is basically I designed the piece(s) to act (such that) this part is easy, the next part is less easy, the next part is the most difficult. So if you consider the entrance to the new CIA building (to be) the entrance to the agency, the

		piece that's out front is the most simplistic, basic code that there is. Then they get more sophisticated the further you (go onto the campus).
Jim Sanborn	Wired News 20 Jan 2005	WN: Do you have to be on the CIA grounds in order to solve Kryptos? Sanborn: No. WN: So just by reading the text taken from Kryptos and posted online, you can solve the puzzle? Sanborn: Well, yeah. That doesn't mean that what I've said in the piece doesn't do something physically there at the agency. So the effect of the piece might affect something at the agency so that you'd have to see what I did at the agency. WN: What do you mean? Sanborn: In part of the code that's been deciphered, I refer to an act that took place when I was at the agency and a location that's on the grounds of the agency. So ... you have to decipher the piece and then go to the agency and find that place. There are, for example, longitude and latitude coordinates on the piece, which refer to locations of the agency. WN: When you say act do you mean an act that you did or that happened while you were there? Sanborn: An act that I could have carried out. I refer to something I'd done out there. WN: Something that you did do? Sanborn: I made reference in the encoded text to something I could have done there.
Jim Sanborn	Wired News 20 Jan 2005	WN: How many (cryptographic) techniques did you use (in Kryptos)? Sanborn: I would think five or six.
Jim Sanborn	Wired News 20 Jan 2005	WN: Are the coded systems you used the same systems that (CIA cryptographer) Ed Scheidt gave you or have they been altered? Sanborn: Mr. Scheidt basically gave me an outline of historic and contemporary ... encoding systems that have been formally used by the agency and were still used by the agency and other people (in 1990). He gave me a whole variety of possible systems to use and ways to modify all of those systems. But as a visual artist, I like to rely on systems that include visual (material) as well as digital material that can be deciphered by machines. It's also well-known that I did use some matrix codes Ed gave me, and I have also designed visual systems for encoding, which are much harder for cryptographers to crack because they're individualistic. WN: But it's still basically code systems that Scheidt gave you, right? I mean he could decipher Kryptos if he wanted to, correct? Sanborn: No, he doesn't know the solution. I made that very clear that I didn't want him to be able to decipher what's going on ... that I'd be modifying systems and

		developing my own, which would make it virtually impossible for him to decipher all of it. I intended the 80 percent (of the text) that's been deciphered to be deciphered and to be deciphered in stages and relatively quickly. The final part is obviously the, you know, the apex of the pyramid there.
Jim Sanborn	Wired News 20 Jan 2005	WN: The Cyrillic Projector, when decrypted, turned out to be text from a classified KGB document instructing agents on how to gather and work with sources. The projector essentially dealt with covert obsolescence, that is, covert activities that became obsolete with the end of the Cold War. Does Kryptos also deal with covert obsolescence? Sanborn: No. Not that it doesn't say something about clandestine trade craft. A lot of the work (I do) deals with secrecy and ... the modus operandi of spies -- how they operate, how they turn sources and things like that. Even the (other) pieces at the agency that are in the courtyard -- stone layers that have encoded text on them -- sort of dealt with secrecy as an entity, which has existed through time for eons and generations. Cave people (for example) keeping secret their methods of killing a mammoth or something like that.
Jim Sanborn	Wired News 20 Jan 2005	WN: If Kryptos is solved, will the answer be something like the Cyrillic Projector's solution -- text from a formerly classified document -- or will it have a larger meaning? Sanborn:: It's a very different animal (from the Cyrillic Projector). The answer will be far more ambiguous. Of the part that's been decoded already there is certain ambiguity in the last few sentences and it's been open to interpretation, as has the whole piece.
Jim Sanborn	Wired News 20 Jan 2005	WN: The Cyrillic Projector has some of the same code that's on the CIA piece, but it does have some alterations. It's not completely identical. Why is that? Sanborn: If you make a body of work over many years so that anyone will know that it's your art, you have a common denominator, which means you leave clues. You don't just sign every artwork. There's something about that particular artwork that you say, "Oh, that's a Sanborn." And (the coding) was one of the common denominators that I chose to leave in. Just like my lodestones. I brought my lodestones forward 10 years and used them at the agency. Then I brought Kryptos forward five years and used them after that. And I will continue to bring work back and carry ideas forward in order to get a continuum for the body of work.
Jim Sanborn	Wired News 20 Jan 2005	WN: Well, you've mentioned before that each night after construction would finish for the day people from the CIA would come out to measure the materials. Sanborn: Well the new (CIA) building was being built while I was there and at night there were teams that used, I believe, a neutron scan (on) everything that went into the agency so that they could find any bugs or anything that had been planted. (They) used ground-penetrating radar and various other means to see and find everything that was there. And I would suppose they did that with my piece as well, which makes it difficult to do whatever you'd like to do -- not in an espionage way, but whatever you want to do.

		WN: What do you mean? Were there things that you wanted to do with the sculpture that you were unable to do? Sanborn: I didn't say I was unable to do it, I just said it makes things difficult. When somebody comes in and X-rays everything you do every night, it makes it tough doesn't it? (Laughs) WN: So you did something, but they knew about it? Sanborn:: No, I didn't say that, either.
Jim Sanborn	Wired News 20 Jan 2005	WN: There is someone who says he thinks he knows how the last section should be decrypted. John Wilson says that section three from Howard Carter's diary is giving instructions to the reader for what they should do with the text in the last section in order to decipher it. For example, when it describes Carter putting the candle through the hole, Wilson says it's an instruction for what to do with the text. So Wilson placed the word candle into the text. Is he on to something or off track? Sanborn: I'm inclined to not comment. If a person deciphers and sends me the exact decipherment -- if it can be deciphered exactly, considering most of my things are rife with mistakes on purpose -- I'd probably let them know that they got it if they did. I will say that I have left instructions in the earlier text that refer to later text. (But) that's as far as I'll go.
Jim Sanborn	Wired News 20 Jan 2005	WN: Do you want the puzzle of Kryptos to be solved? Sanborn: Uhhh ... I certainly want it to be considered. I had figured that the parts that have been solved already would have been done a lot quicker than they were. But that might just have been a question of focus of the cryptography community.
Jim Sanborn	Wired News 20 Jan 2005	WN: One decoded section refers to a "WW." It says specifically, "Who knows the exact location? Only WW." Who is WW? Sanborn: (Former CIA Director) William Webster.
Jim Sanborn	Wired News 20 Jan 2005	WN: The CIA required you to write down the solution to the sculpture and give it to Webster. Webster has said that he forgot the solution. Did he ever actually know the answer? Sanborn: Well, you know, I wasn't completely truthful with the man. And I'm sure he realizes that. I mean that's part of tradecraft, isn't it? Deception is everywhere. I had to leave an envelope at the agency saying what was on the (sculpture). I gave it to William Webster at the dedication ceremony with a wax seal on it, but in fact I really didn't tell him the whole story. I definitely didn't give him the last section, which has never been deciphered. WN: But he thinks you did give him the solution. Sanborn: That's his problem! (laughs)
Jim Sanborn	Wired News 20 Jan 2005	WN: Do you remember what the solution is? Sanborn: No.

		[NOTE: This later changed, by the time he released the "BERLIN" clue -RSP] WN: You don't remember the solution to your own sculpture? Sanborn: No. I've got it hidden someplace but I'm not going to read it. I have done everything I can to forget (it). Because I don't want to slip and give somebody information about it. I mean, you read the piece of paper, you burn it, and you forget it. That's the only way information is kept secret. (Otherwise) it's very difficult not to give clues. In the early days, anything I said was a clue. Now things are getting more and more refined the more people (are looking into) this. They are looking for shreds, tiny little slivers of information. So I have to be very careful not to go any further. WN: So if you don't know the answer how will you know if anyone has solved it? Sanborn: I have the solution hidden someplace. So if somebody cracks it I can cross-check it. WN: What if something happens to you? Sanborn: The secret will probably pass away. WN: You haven't left it in your will? Sanborn:: Well, actually I have. I think it's important that whoever says that they cracked it will in fact find out whether they actually did. So from that standpoint, there does have to be some sort of historic record of what it says.
Jim Sanborn	ABC 1991	What's the title of this piece, and what does it mean? The title is Kryptos, and it's significant in a couple of ways. It's the Greek term for "hidden". And I've always worked with invisible forces, or making forces visible that are invisible, or visa-versa, and make visible things invisible. And so I chose that word as my keyword for the code also. I did it in large text that's cut through a copperplate out at the Agency. And that -- the keyword for some of that text is Kryptos.
Jim Sanborn	ABC 1991	Does that for you symbolize the CIA? Well, to tell you the truth, when I was first commissioned, my ideas of what the Agency did and what the Agency was were quite different... What did you think it was? Um, it was spies and spying. And it's something different? It's really information gathering. It gathers information, processes it, and then transfers that information, and people use the information. That's its primary, that's what I understand now, is its primary mission. Obviously there are other layers, but those are the ones that I worked with, and those are the ones that I understand now.

		That information is what I – see all the information comes in, encoded generally in some form, and then it's processed and then sent out, encoded, in secret.
Jim Sanborn	ABC 1991	Well did you say to yourself, "Hmm, let me see if I can fool them with this?" Sure. Yes. I mean, I ... The levels of complexity developed while I was building the piece. But I always thought that I wanted to fool them. Y'know that was always the plan from the very beginning. I don't know – fool them or make everybody out there think a little harder about maybe it isn't just things military which can be secret, or things like that. I mean it's a more general statement about secrecy that I make.
Jim Sanborn	ABC 1991	How did you decide what message to encode? Well that changed during the course of the project as well. And that developed largely from the reading that I did about the Agency. And also while I was cutting out, physically cutting out the copperplates, things in my own mind changed. I wanted to put something on there which would -- Let's just say, that once the plate is deciphered, I'm not convinced the true meaning will be clear even then. I've made a statement which is straightforward, but that leads to something else. There's another deeper mystery. As you peel off a layer of an onion, the myst-- you get closer to the heart of what it is. And so y'know I just wanted to make it even after it was deciphered, you had to go deeper and decipher something else. It's in English, it's in plain English, it's in text, and you can read it, but that isn't necessarily the whole story.
Jim Sanborn	ABC 1991	How about a little? How about a little? Without, without... Just a context, I mean, so that somebody sitting at home would say, "Well, what are they talking about?" Is there.... Is this a famous quotation, for example? I can't say whether it's a quotation. It is either a quotation or it's something I've written, or it's something somebody else has written. But let's just say, that it has a lot to do with secrecy. It has a lot to do with finding something, or opening something, or discovering something that's secret.
Jim Sanborn	ABC 1991	What have you heard from people out there about this? Of course, it's indirect information, I don't necessarily have an inside person. I hear it by the grapevine or friends of friends. It's my understanding that the simpler parts of the encoded texts have been deciphered, as they had planned to be. Part of it was designed to be deciphered within a couple of weeks. Part of it was designed to be deciphered within a couple of months. And part of it within years, or never.
Jim Sanborn	Letter to CIA Employees 15 Dec 1989	Second, the tilted strata tell a story like pages of a document. Over the next several months, a flat copper sheet through which letters and symbols are cut will be inserted between these stone "pages". This code, which includes certain ancient ciphers, begins as International Morse and increases in complexity as you move through the piece at the entrance and into the courtyard.
Jim Sanborn	Letter to CIA Employees 15 Dec 1989	The right side is a text that can be partly deciphered by using the table and partly by using a potentially challenging encoding system. The text, written in collaboration with a prominent fiction writer, is revealed only after the code is deciphered. <i>NOTE: I believe it was later stated that the fiction writer was not used.</i>

Jim Sanborn	Letter to CIA Employees 15 Dec 1989	My choice of materials, like code, conveys meaning. At the entrance, a lodestone (a rock naturally magnetized by lighting [sic]) refers to ancient navigational compasses. The petrified tree recalls the trees that once stood on this site and that were the source of materials on which written language has been recorded. The copper, perforated by text, represents this 'paper'.
Jim Sanborn	New York Times 21 Nov 2010	"I assumed the code would be cracked in a fairly short time"
Jim Sanborn (via reporter)	New York Times 21 Nov 2010	Mr. Sanborn admitted to introducing misspellings to add a degree of difficulty.
Jim Sanborn (via reporter)	New York Times 21 Nov 2010	Mr. Sanborn has said the passage [King Tut quote in K3] has inspired him since childhood.
Jim Sanborn	New York Times 21 Nov 2010	Anybody holding a secret has a position of power, even if it's a trivial secret.
Jim Sanborn (via reporter)	New York Times 21 Nov 2010	He has asked a friend — Mr. Sanborn has given him some of the answers, but not all — to handle inquiries from those who say they have solved the puzzle. "My friend says, 'What do you have in letter-position 27?' " Mr. Sanborn explained. "If that's not the same letter, 'Game over, and you didn't crack the code.'"
Jim Sanborn	Wired 20 Apr 2009	The whole thing is about the power of secrecy.
Jim Sanborn	Wired 20 Apr 2009	The text was scrambled by "a coding system that would unravel itself slowly over a period of time."
Ed Scheidt	Wired 20 Apr 2009	I was reminded of my need to preserve the agency's secrets... You know, don't tell him the current way of doing business. And don't create something that you cannot break—but at the same time, make it something that will last a while.
Jim Sanborn	Wired 20 Apr 2009	I assumed the first three sections would be deciphered in a matter of weeks, perhaps months.
Ed Scheidt (via reporter)	Wired 20 Apr 2009	Scheidt figured the whole puzzle would be solved in less than seven years.
Jim Sanborn	Wired 20 Apr 2009	I tried to make [K1] sound good and be inscrutable enough to be interesting,

Jim Sanborn and Ed Scheidt (via reporter)	Wired 20 Apr 2009	Both Scheidt and Sanborn confirm that they intended the final segment to be the biggest challenge.
Jim Sanborn	Wired 20 Apr 2009	It's not my intent to put out disinformation... I'm a benevolent cryptographer.
Jim Sanborn	Wired 20 Apr 2009	[regarding the plaintext, and whether William Webster has it] Nobody has it all, I tricked them.
Jim Sanborn	Wired 20 Apr 2009	If somebody tried to torture me, I couldn't tell them... I haven't looked at the plaintext of K4 in a long time, and I don't have a very good memory, so I don't really know what it says.
Ed Scheidt	Wired 20 Apr 2009	There may be more to the puzzle than what you see... Just because you broke it doesn't mean you have the answer.
Jim Sanborn	Wired 20 Apr 2009	In some ways, I'd rather die knowing it wasn't cracked... Once an artwork loses its mystery, it's lost a lot
Jim Sanborn (via reporter)	New York Times 16 Jun 1999	Mr. Sanborn said this week that the sculpture contains a riddle within a riddle -- one that will be solvable only after the four encrypted passages are known.
William Webster	New York Times 16 Jun 1999	I have zero memory of this [answer to Kryptos]... It was philosophical and obscure.
Jim Sanborn (via reporter)	New York Times 16 Jun 1999	Mr. Sanborn... has said he believed that the ultimate secret hidden in the text of 'Kryptos' will never be deciphered.
Ed Scheidt	Wired 21 Jan 2005	I provided the cryptographic process as well as worked with him with what he was looking to do as far as the story (the sculpture would tell). We came up with a methodology using some of the known cryptographic solutions (at the time).
Ed Scheidt	Wired 21 Jan 2005	Wired: How many cryptographic processes are used on the sculpture? Scheidt: There are four different processes. Two of them are similar and the other two are different things. The first three processes were designed so that a person could, through cryptographic analysis, have access to the English language (on the sculpture). And the last process, I masked the English language so it's more of a challenge now. It's progressively harder in the challenges. [Note: when Scheidt refers to 4 'processes', I believe he is referring to K1, K2, K3, and K4, not cryptographic techniques]

Ed Scheidt	Wired 21 Jan 2005	All four (sections) are done in the English language. The message could have been in another language. (But) this particular puzzle is in the English language.... The techniques of the first three parts, which some people have broken, (used) frequency counting and other techniques that are similar to that. You can get insight into the sculpture through that technique because the English language is still visible through the code. (But with) this other technique (in the fourth part), I disguise that. So ... you need to solve the technique first and then go for the puzzle.
Ed Scheidt	Wired 21 Jan 2005	Wired: Are all four of the processes known processes? There aren't any that aren't known, right? Scheidt: The masking technique may not be known.
Ed Scheidt	Wired 21 Jan 2005	Wired: Jim said that he took your techniques and then he deliberately masked them even more so that even you wouldn't know what was in the puzzle. Scheidt: Ah hah. I can't respond to that one. I haven't heard him say that before. It's possible I guess. I haven't talked to Jim on what he did, but I do know ... there was some masking techniques that were used and that's about it.
Ed Scheidt	Wired 21 Jan 2005	Wired: So you may not know what the message says. Scheidt: Obviously there could be a possibility. I know what the message was to be. (But) since he's the one who had the chisel in his hands, there could be some changes.
Ed Scheidt	Wired 21 Jan 2005	Wired: But within the text that's already been translated it does mention something being buried on the grounds of the CIA. Scheidt: Well, the idea of encoding a message is not only to encode the externals of a message (the English language), but the message itself. Once it is readable, it may have other encoding that's involved in it. That's something that would show up in secret messages. If I wanted to, for instance, say (that) you and I are going to meet at 1 o'clock on Friday. We may establish a code that 1 o'clock on Friday is equal to "cake." So in my message I would say how about you and I meeting at a convenient place for cake? Then you and I really know that cake means the time. [This suggests that nothing is physically buried, and that K2 may be readable but with more encoding...]
Ed Scheidt	Wired 21 Jan 2005	Wired: So someone could translate the actual message but not know what the message means. Scheidt: That's right. And that's where the masking and all these other kinds of techniques can come into play.
Ed Scheidt	Wired 21 Jan 2005	It's a question of how would you approach it? Would you approach it mathematically? Would you approach it in the context of secret writings or symbols? There's a whole array of things which offer a challenge.
Ed Scheidt	Wired 21 Jan 2005	Another way of saying it is that as one peels the onion away, or the various cloaks, you come closer to the truth from a philosophical sense. And then where does it take you down that path? Well I would assume that the path will lead you down different ways, depending on your philosophical perspective. So it's again back to is it a

black-and-white answer or is it an answer that has a lot of gray areas?

Sources:

NPR	09 Jun 2005 NPR "All Things Considered".
Wired.	20 Jan 2005 Partial transcript of a Wired News interview by Kim Zetter. No longer online. It was edited "for length and organization."
1991a.pdf	Transcript of B-roll interview between ABC's John Martin and Jim Sanborn, 1991. [Video at http://www.youtube.com/watch?v=TT_MamKYits]
(page 8)	Letter from Jim Sanborn to CIA employees.
New York Times	20 Nov 2010 "Clues to Stubborn Secret in C.I.A.'s Backyard"
Wired	20 Apr 2009
New York Times	16 Jun 1999
Wired	21 Jan 2005 Ed Scheidt interview

TIMELINE:

- November 1988 - Sanborn chosen to create Kryptos sculpture
- November 1990 - Kryptos dedication ceremony
- November 1992 - NSA decoded K1-K3 within 2 days.
- February 1998 - David Stein revealed his partial solution at Langley, but kept secret
- June 1999 - Jim Gillogly cracked K1-K3.
- April 2006 - Sanborn states that "IDBYROWS" was incorrect (it should have been "XLAYERTWO")
- 21 Nov 2010 - Berlin clue released
- 2010 - Sanborn released the Kryptos replica